FEEL BEFORE YOU BUILD

Hands-on prototyping with actuators · 2 hours

AARHUS
UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE

MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

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WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Today you put a vibration motor against your own wrist, decide it feels cheap, and pick something else. Better now than in week 46.

Every year, projects end up built entirely on the screen and the speaker.

Not because touch was the wrong choice. Because nobody knew that a coin motor costs 12 kr and takes two wires.

HOW TODAY WORKS

Six benches. Already wired. Already running.

• Ten minutes each, on a hard timer.

• You change one parameter and answer one question.

• You wire nothing. That is what the project weeks are for.

Then you build one signal and someone else tries to read it.

RUN SHEET

Times are offsets from the start. The rotation is the spine of it, so keep it moving.

0:00	10 min	Two motors, same message. Which felt urgent?
0:10	60 min	Bench rotation, 10 minutes each
1:10	10 min	Round-the-room: what surprised you
1:20	30 min	Build one signal
1:50	10 min	Blind test and pack down

THE SIX BENCHES

01 ERM COIN MOTOR

CUTANEOUS · VIBRATION

An off-centre weight on a motor shaft. The same part that is in your phone and every game controller.

Turn the knob and notice what you cannot do. Speed and strength are mechanically welded together, so "gentle but fast" is not available to you.

```
part     10mm 3V coin ERM
drive    2N2222 + 1N4148
pin      D9 (PWM)
draw     ~75 mA
spin-up  20–40 ms
cost     ~12 kr
```

CARD QUESTION

At what point does it stop feeling like information and start feeling like a malfunction?

02 LRA + PIEZO DISC

CUTANEOUS · VIBRATION

Two ways out of the ERM's compromise.

The LRA hits resonance in about 5 ms and stops just as fast, which is why phone keyboards use it.

The piezo is a crystal that flexes when you put current across it. Near-instant, almost no power, but shallow. A tick, not a thump.

```
parts    LRA 10mm + piezo
driver   DRV2605L (I2C)
pin   A4 SDA / A5 SCL
effects   123 waveforms
cost   ~90 kr
```

CARD QUESTION

Which of the three vibrators would you actually put on a wrist, and why not the others?

03 SOLENOID TAP

CUTANEOUS · IMPACT

A push-pull solenoid firing a single 15 ms pulse against a fingertip.

This is the bench everyone remembers. A single knock carries urgency that no amount of buzzing does, because it reads as a person tapping you rather than a machine signalling.

```
part    5V push-pull
drive   MOSFET + flyback
pin     D6
peak    ~1.1 A
duty    <= 25%, gets hot
cost    ~45 kr
```

CARD QUESTION

How many taps before it goes from alert to nagging?

04 CAPACITIVE TOUCH

KINESTHETIC · TOUCH-TO-ACTUATE

Touch a surface with a bare finger and the servos react.

There is no button and no sensor you can point at. The input is a wire taped behind a plate, read on an analogue pin.

Any surface can become an input this way: foil, card, fabric.

```
sense     ADCTouch on A0
baseline  25-sample roll
trigger   1.01x average
debounce  100 ms
servos    D4 + D13
cost      ~0 kr, a wire
```

CARD QUESTION

Put your hand near the plate without touching. When exactly did it decide that was a touch?

BENCH 04: THE WHOLE TRICK

Touch input for the price of a wire. The thinking is where it costs you.

reading > baselineAverage() * 1.01

The threshold is RELATIVE, never absolute.

• Readings drift with humidity, with mains hum, with how you are sitting. A fixed threshold works in the morning and fails after lunch.

The baseline only learns while untouched.

• Otherwise a long press slowly teaches it that a finger is normal, and the touch quietly stops registering.

05 PELTIER WARM / COOL

CUTANEOUS · THERMAL

A thermoelectric tile: heats one side, cools the other, reverses when you flip the current.

Sit with it. It takes several seconds before you are sure which way it went.

And alternating hot with cold does not read as a pattern. It just reads as lukewarm.

```
part    TEC1-12706
drive   L298N H-bridge
pin     D3 PWM / D4-5
draw    2–4 A, bench PSU
onset   3–8 s
cap     45 C in software
```

CARD QUESTION

Time yourself. How long until you would bet money on warmer vs cooler?

06 THE SAME SIGNAL, TWICE

INTRAMODAL · AUDIO + HAPTIC

One transducer on a plate, playing 40 Hz you feel and 400 Hz you hear, both from the same driver.

Toggle between them alone and together.

Together is not louder. It is more certain. Sending the same thing down two channels is the cheapest reliability your project can buy.

```
part    bone-conduction
amp    PAM8403 class-D
pin    D11 (tone)
felt    ~40 Hz
heard    ~400 Hz
cost    ~110 kr
```

CARD QUESTION

With ear defenders on, does the signal still work?

INPUTS, AND PRIOR ART

INPUTS WORTH KNOWING ABOUT

One table, not a rotation. Wander over during the build.

- Capacitive touch. A wire on an analogue pin. See bench 04.
- Heart rate (PPG). Steady at rest, useless once the hand moves.
- Skin conductance (GSR). Drifts constantly, so relative change only.
- Flex sensor. The cheap route to a data glove.
- Pressure (FSR). How hard, not just whether. Calibrate each pad.
- Your own phone. Accelerometer, gyro, mic, camera, vibration motor. No soldering at all.

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Capacitive sensing, servos and actuators (Arduino Uno)

• A moving screen that breathes when idle and retreats when you touch it, with a browser control panel over the serial port.

• Take away: easing and idle motion are what separate "a servo moved" from "it reacted to me". Both are a dozen lines.

Instrumented sock (ESP32, bachelor project)

• Six force sensors under a foot, sent to a server over Wi-Fi in batches, each with its own calibration constants.

• Take away: calibrate per sensor, and batch your writes.

BUILD ONE SIGNAL

THE BRIEF

30 minutes. Deliberately not long enough to be precious about it.

Pick one actuator from the rotation.

Encode three messages: arrived, something wrong, finished.

Vary only two things: the RHYTHM of the pulses, and how STRONG they are. No swapping actuators between messages.

Hand it to another group with the screen turned away. They name all three.

Write down which two got confused, and what would have separated them.

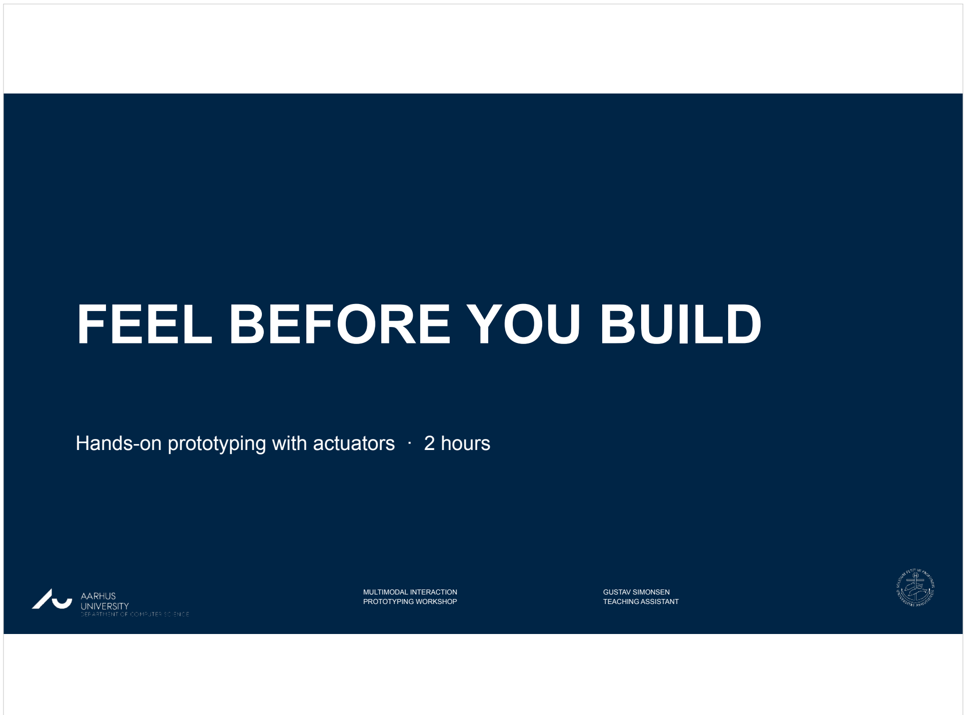
THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If your design only works when someone is watching a screen, you built a visual interface with a motor glued to it.

GO AND TOUCH THINGS

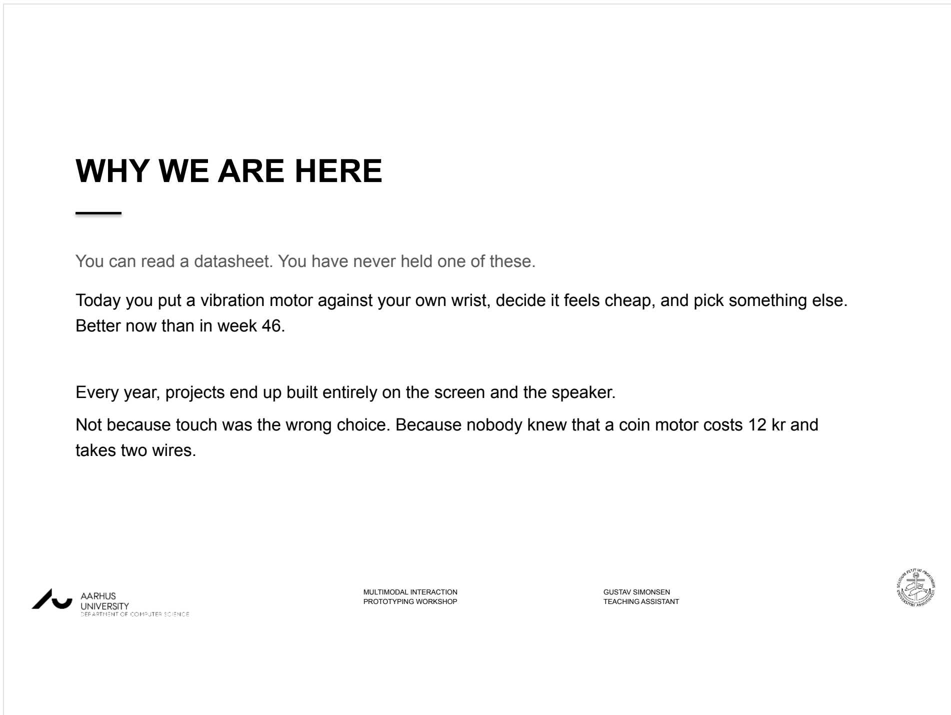
github.com/gust1527/multimodal-actuator-workshop



0:00. Start here. Don't read the slide.

Open with the failing demo, not with talking. Have the coin motor and the solenoid both wired and ready before anyone sits down.

Buzz a phone-style notification on the ERM. Then send the same message on the solenoid. Ask: which felt more urgent? Let them answer. Nobody needs theory for that, and that IS the argument for the whole session.



Say the honest version: every year projects end up on the screen and the speaker, and touch gets designed on paper then dropped.

Not because it was wrong. Because nobody had held the parts.

They get the Actuators lecture tomorrow. Today is the hands, not the theory.

Keep this short, 2 minutes. They came to do things, not hear framing.

HOW TODAY WORKS

Six benches. Already wired. Already running.

• Ten minutes each, on a hard timer.
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Then you build one signal and someone else tries to read it.

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Set the rules firmly, because the rotation collapses if you don't.

Nobody wires anything today. Everything is pre-wired and running. They turn one knob and answer one question.

Say out loud: the timer is hard. Ten minutes means ten minutes. Bench 3, the solenoid, is the one people will not want to leave.

RUN SHEET

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Show it, don't narrate it. Fifteen seconds.

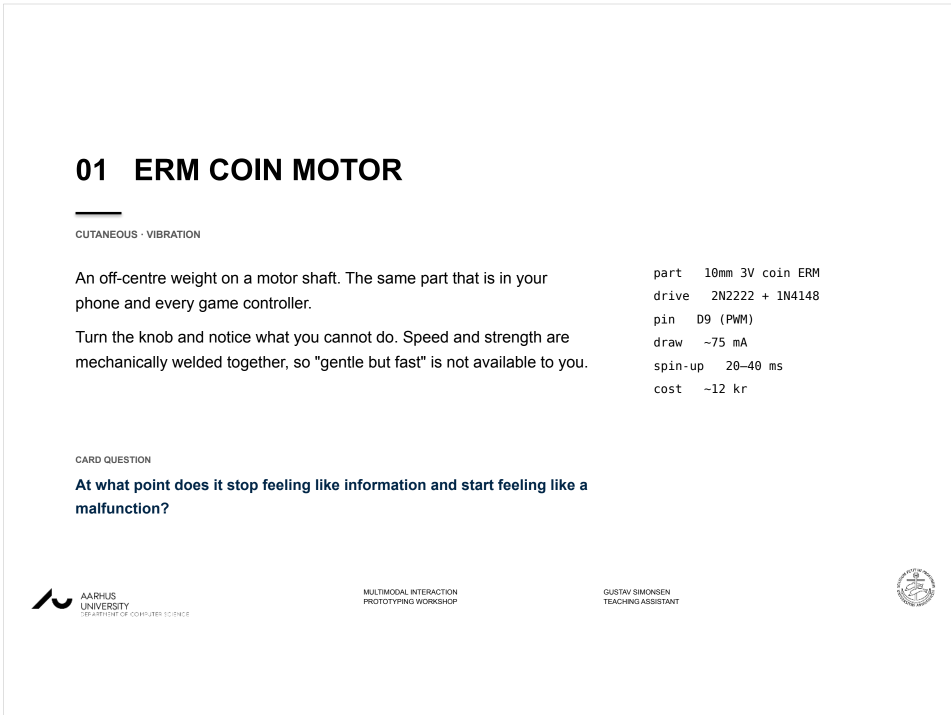
Practical: assign each group a DIFFERENT starting bench so they rotate cleanly.
Six groups, six benches. Write starting numbers on the whiteboard.

If a bench breaks mid-session, pull it and go to five. Five working benches
beat six with a mystery.

0:10. Rotation starts. Set a visible timer.

You are now a timekeeper, not a lecturer. Call the rotation loudly every ten minutes. Circulate, but do not rescue. Let them poke at things.

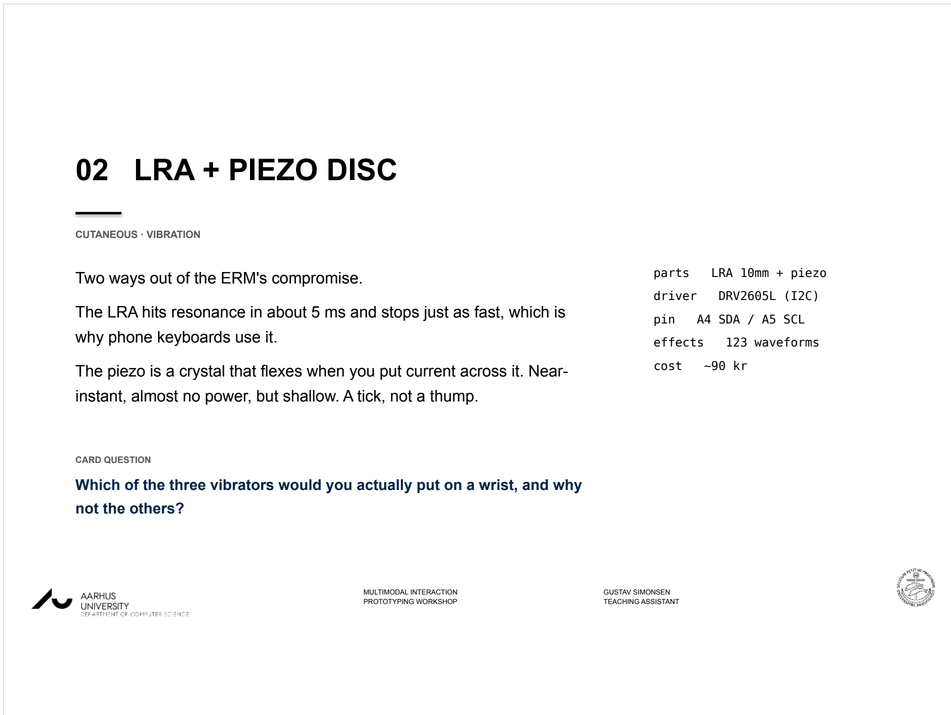
The bench slides that follow are reference for YOU and for anyone who wants to look back. Do not present them one by one before the rotation. That would eat the hour.



The core limitation, made physical: speed and strength are mechanically
linked on this motor. You cannot get gentle-but-fast.

Common question: "why does it feel weak at low PWM?" Below about 90 the motor cannot overcome its own friction. That floor is real and worth naming.

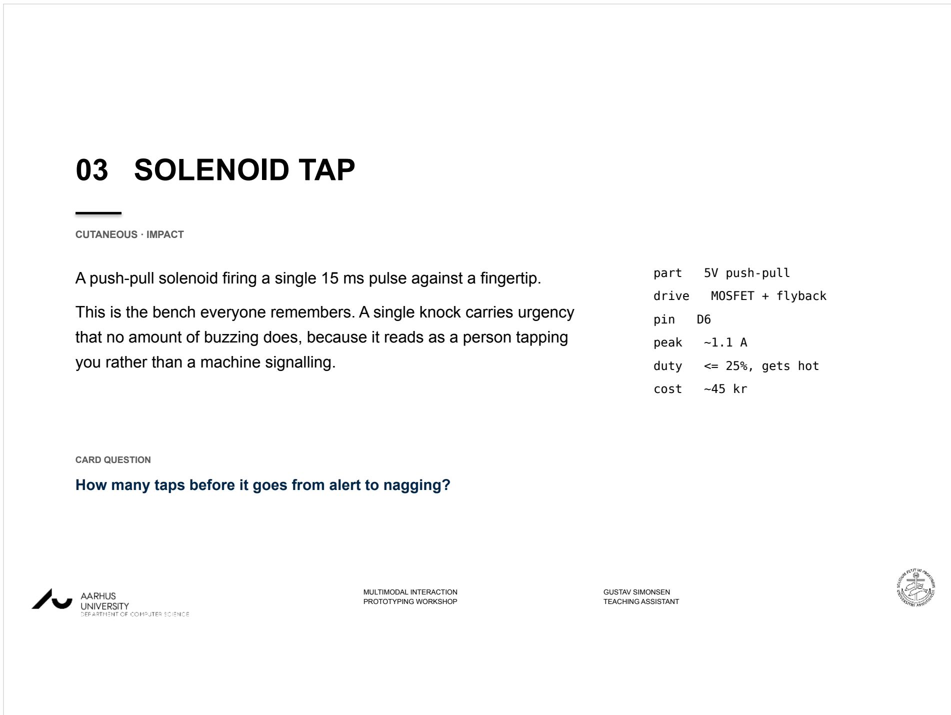
Never let anyone rewire this to a bare pin. 40 mA limit, motor pulls 75.



The escape from bench 1's compromise. LRA is the phone-keyboard feel:
5 ms on, 5 ms off. Piezo is sharp but shallow.

If the LRA sounds wrong, the DRV2605L is in ERM mode. It must be told which type it's driving. That's the failure to expect here.

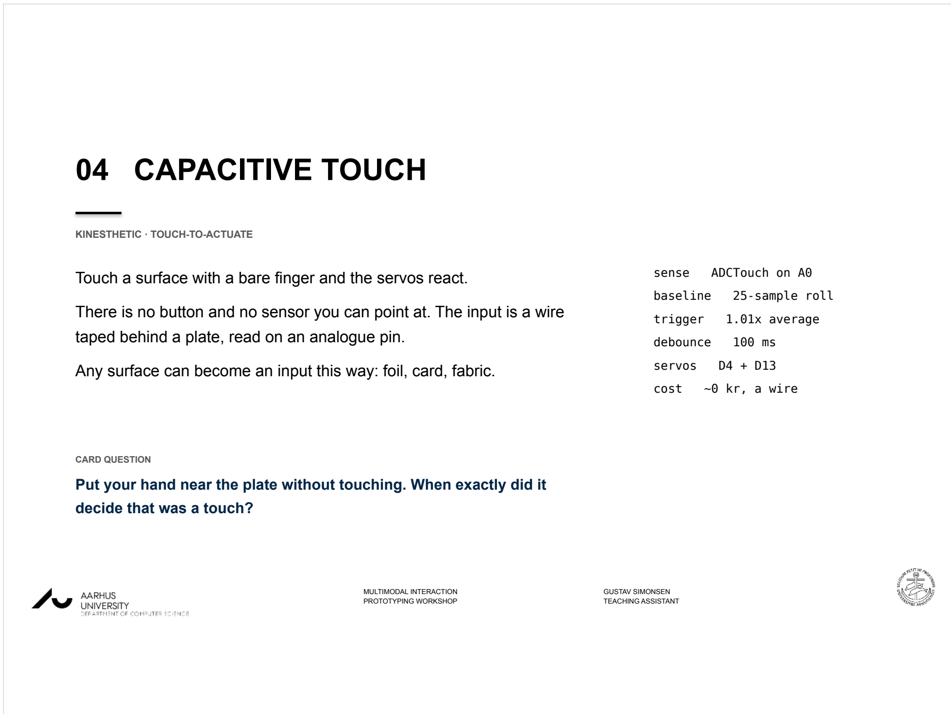
Good moment to point out: 123 built-in waveforms means they don't have to design vibration from scratch in their project.



The memorable one. A single knock reads as a person tapping you
rather than a machine signalling. Worth naming out loud, because it is the
whole reason it feels different.

Watch the duty cycle. Solenoids get hot fast, so if it has been hammering for
ten minutes, let it rest.

Separate supply, common ground. On USB alone the board browns out and resets,
which looks like a code bug and isn't.



The surprise bench. No breakout board. The input is a wire.

Have the Serial Plotter on a screen here. Watching the baseline wander while the trigger line tracks it teaches the whole idea in ten seconds.

Let someone hold their hand NEAR without touching. The gradual approach is what makes the threshold question real.

BENCH 04: THE WHOLE TRICK

Touch input for the price of a wire. The thinking is where it costs you.

reading > baselineAverage() * 1.01

The threshold is RELATIVE, never absolute.

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The baseline only learns while untouched.

• Otherwise a long press slowly teaches it that a finger is normal, and the touch quietly stops registering.

Slow down here. This is the one idea most worth stealing from today.

Two things students get wrong:

1. They use a fixed threshold. Works in the morning, fails after lunch,
   because the reading drifts with humidity and mains hum.

2. They feed the baseline buffer always. Then a long press teaches the
   baseline that a finger is normal, and the touch silently stops working.
   The buffer must ONLY learn while untouched.

05 PELTIER WARM / COOL

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A thermoelectric tile: heats one side, cools the other, reverses when you flip the current.

Sit with it. It takes several seconds before you are sure which way it went.

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draw   2–4 A, bench PSU
onset   3–8 s
cap   45 C in software
```

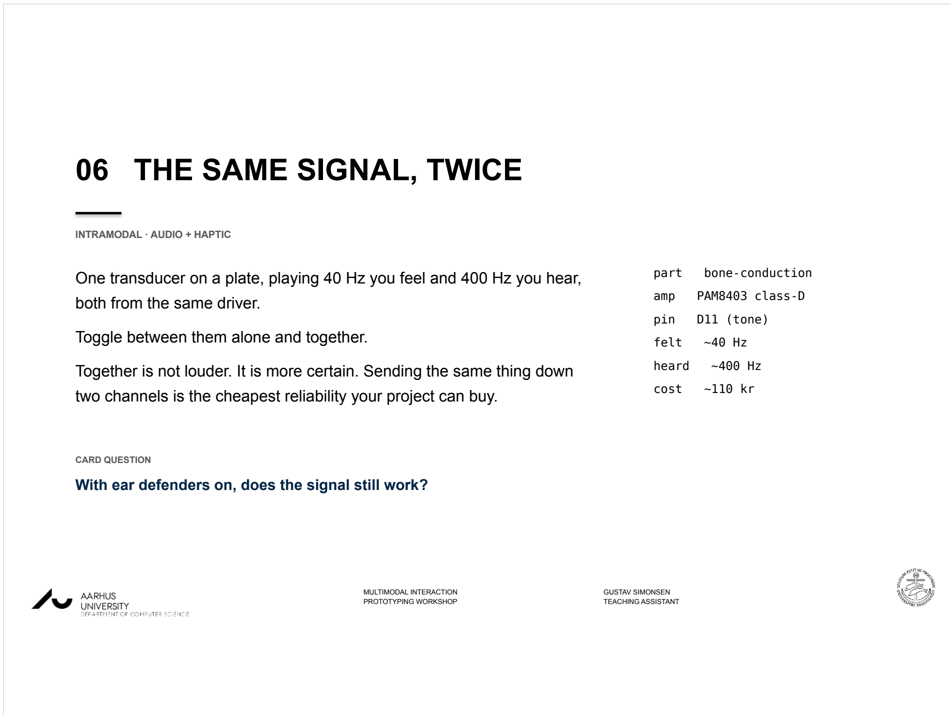
CARD QUESTION

Time yourself. How long until you would bet money on warmer vs cooler?

Thermal. People are bad at reading thermal patterns, and this is
where they find that out for themselves.

Make them time themselves. The 3 to 8 second onset kills a lot of bad project ideas before they get proposed, which is exactly what you want happening now rather than in week 46.

Safety: heatsink on the hot side, capped at 45 C. Skin burns above 50.



Redundancy, made physical. Same driver, two frequencies: one you feel, one you hear.

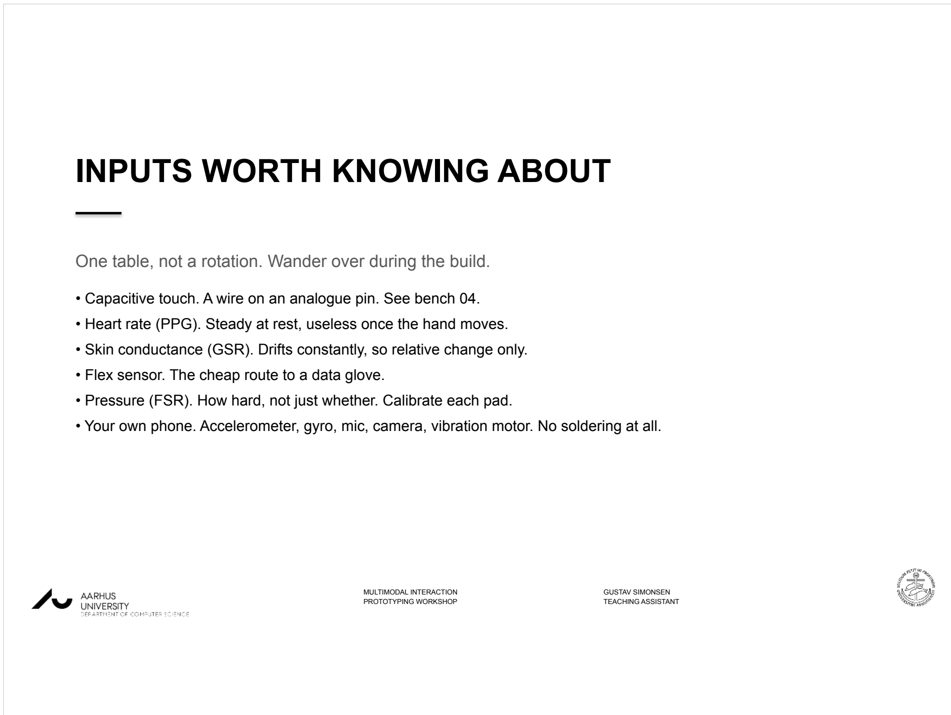
The point isn't that together is louder. It's that together is more CERTAIN. Sending the same thing down two channels is the cheapest reliability their projects can buy.

Ear defenders must actually be used or the comparison is dishonest.

1:10. Rotation over. Round-the-room now.

One sentence per group: what surprised you.
No slides, no laptops. Ten minutes.

Expect the thermal group to report the delay.
Let that land, it is more
persuasive coming from a peer than from you.



Mention, don't teach. These sit on one table with a sign.

The honest warnings matter more than the specs: PPG is garbage once the hand moves, GSR drifts so only relative change is usable, FSRs need per-pad calibration.

Push the phone option hard. For a lot of projects it genuinely beats everything else on the table and needs no soldering.

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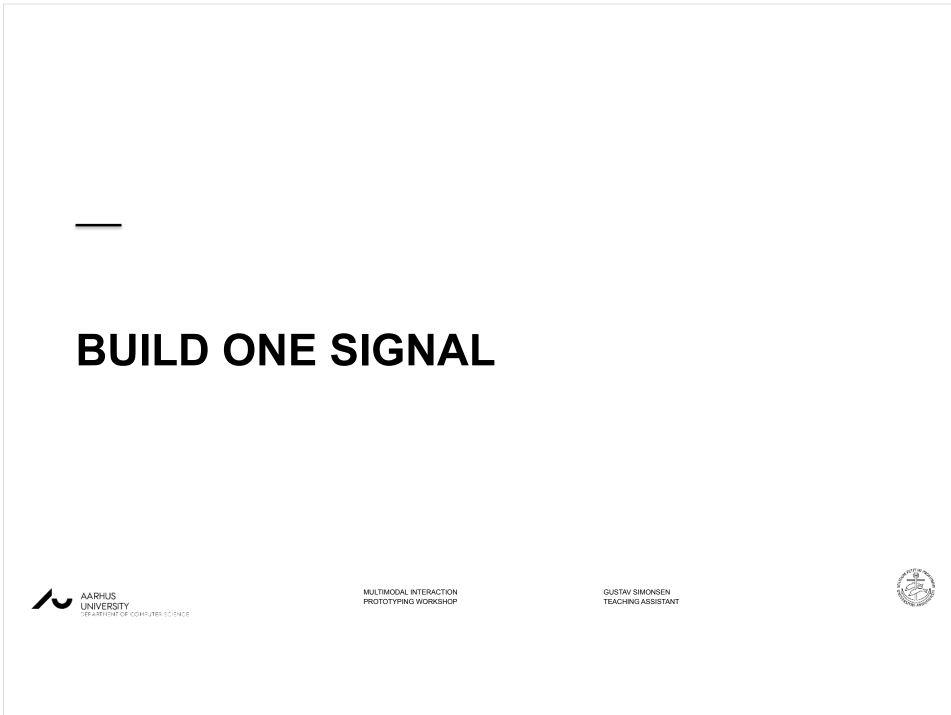
- Six force sensors under a foot, sent to a server over Wi-Fi in batches, each with its own calibration constants.
- Take away: calibrate per sensor, and batch your writes.

Five minutes, and show the actual repos on screen if you can.

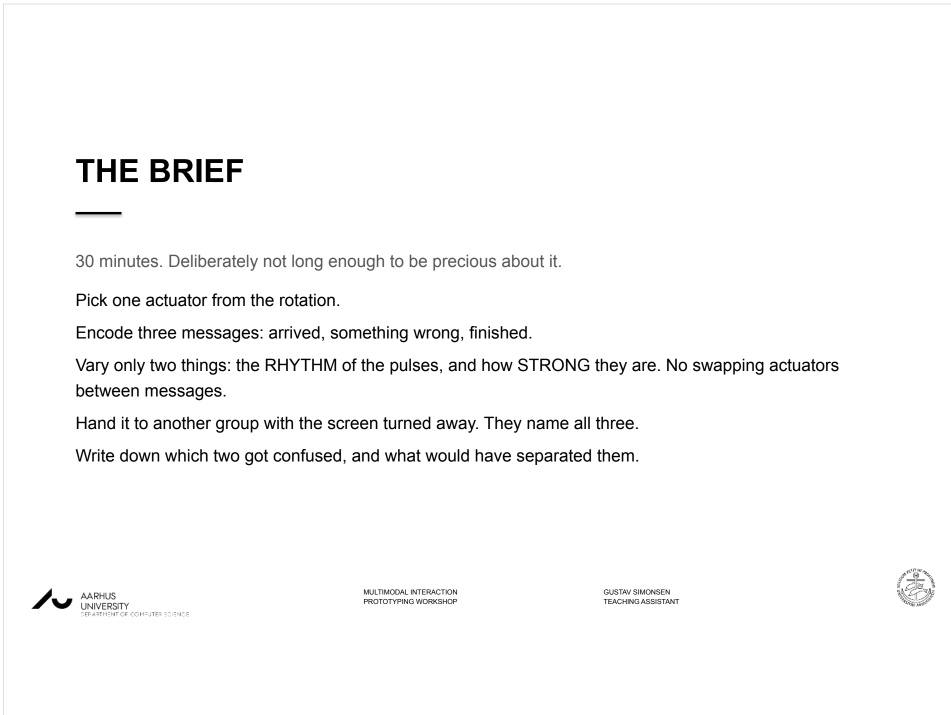
The argument: this was built by students at your stage, not by a lab with a budget. The gap between "we should use haptics" and "it moves when you touch it" is smaller than it looks.

Note for you: the sock repo has a hardcoded Wi-Fi password and live patient IDs in it. Scrub before showing, or show only the calibration section.



1:20. Build starts. Thirty minutes.

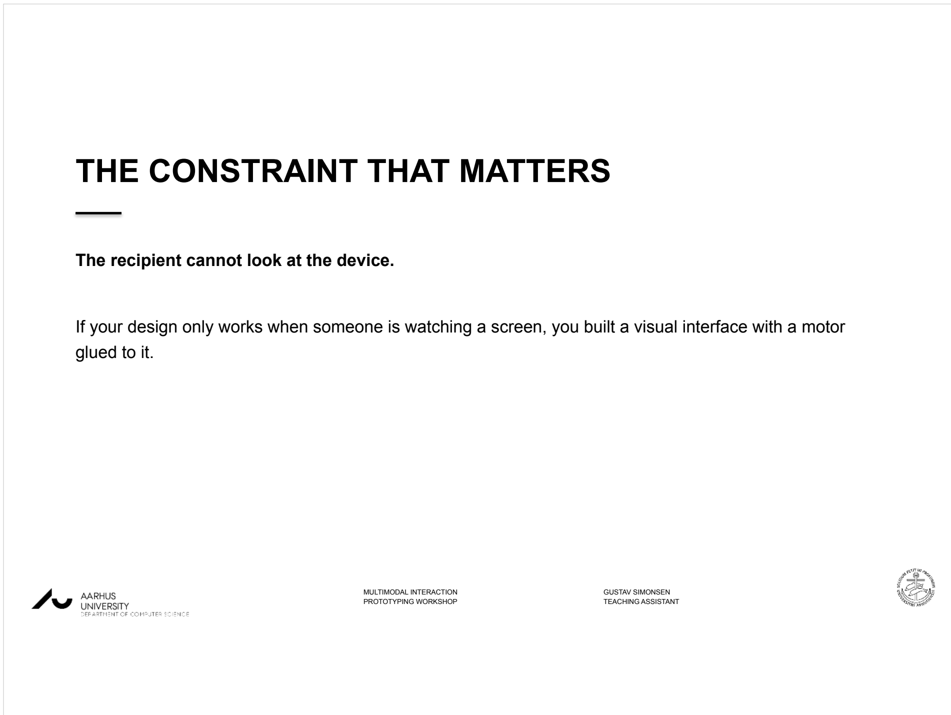
Groups pick any actuator from the rotation. Most will pick bench 1 or 3
because they are simplest. That is fine, the constraint does the work.



Read the brief out once, then get out of the way.

Rhythm and strength only. If someone asks to add a second actuator, say no.
The constraint is the exercise. Encoding within one channel is the skill.

Suggested set is on the slide, but let them choose their own three messages if
they have a better idea.



This is the line worth repeating twice.

1:50. Blind test. Swap groups, receiver looks away, guess all three.

Then have each group name which two got confused and what would have separated them. That confusion is the actual learning, not the successful ones.

Pack down: everything back in the box, including the resistors. It will not all come back.

Point them at the repo. Bench 01 and 04 sketches are commented for them.

Last thing to say: pick your actuator before you design the interaction, not
after. Today was so they can do that from experience rather than a datasheet.

If anyone wants to borrow parts for the project weeks, now is when they ask.